

In [1]: `import matplotlib.pyplot as plt`

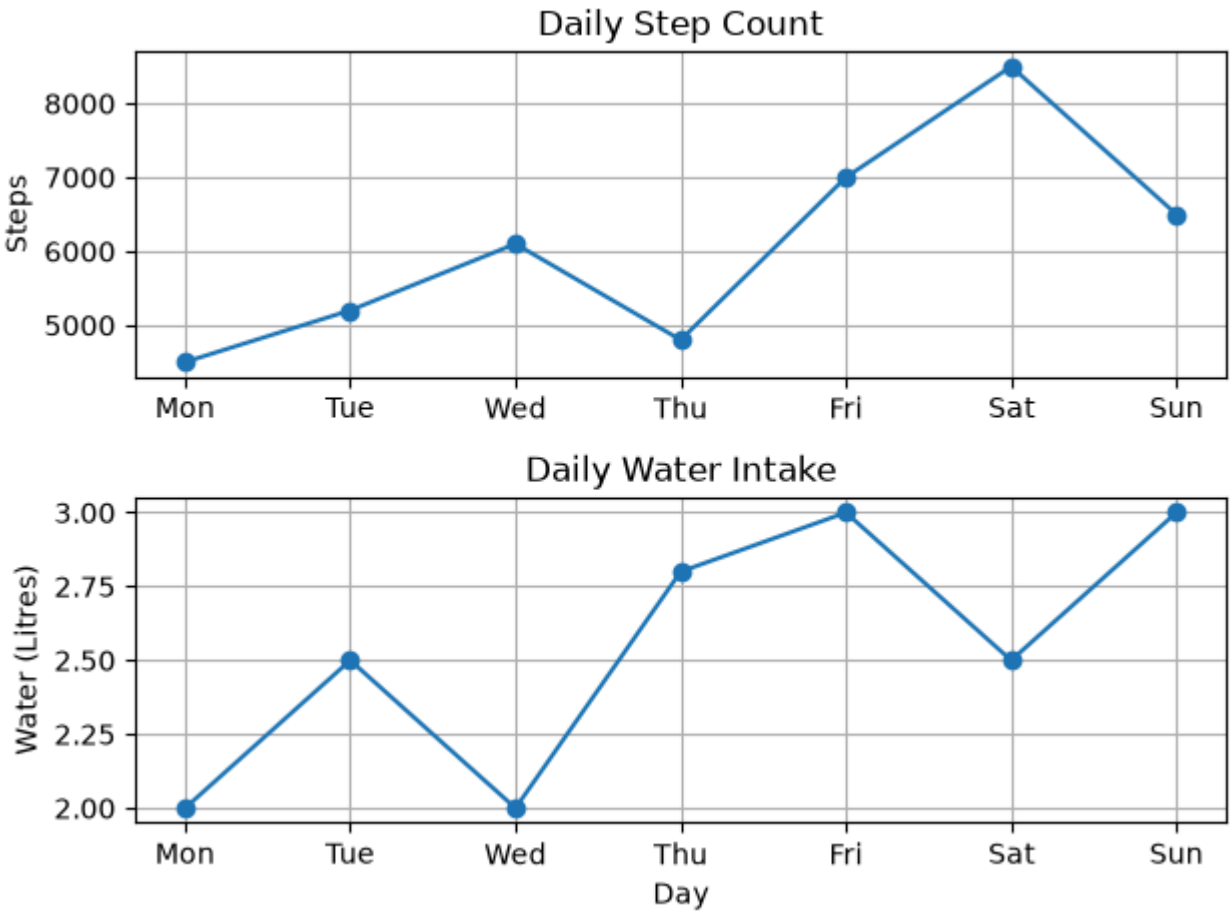
In [3]: `#Task 1 – Daily Steps & Water Intake`

```
days = ["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"]
steps = [4500, 5200, 6100, 4800, 7000, 8500, 6500]
water = [2, 2.5, 2, 2.8, 3, 2.5, 3]
```

```
plt.subplot(2, 1, 1)
plt.plot(days, steps, marker="o")
plt.title("Daily Step Count")
plt.ylabel("Steps")
plt.grid()
```

```
plt.subplot(2, 1, 2)
plt.plot(days, water, marker="o")
plt.title("Daily Water Intake")
plt.xlabel("Day")
plt.ylabel("Water (Litres)")
plt.grid()
```

```
plt.tight_layout()
plt.show()
```



In [5]: `# Task 2 – Instagram Posts by Quarter`

```
quarters = ["Q1", "Q2", "Q3", "Q4"]
```

```
posts = [25, 32, 28, 40]
fig, ax = plt.subplots(2, 2, figsize=(10, 8))
```

```
#Q1 Line
ax[0, 0].plot(quarters, posts, marker="o")
ax[0, 0].set_title("Instagram Posts - Line Chart")
ax[0, 0].set_xlabel("Quarter")
ax[0, 0].set_ylabel("Number of Posts")
ax[0, 0].grid()
```

```
# Q2 - Bar chart
```

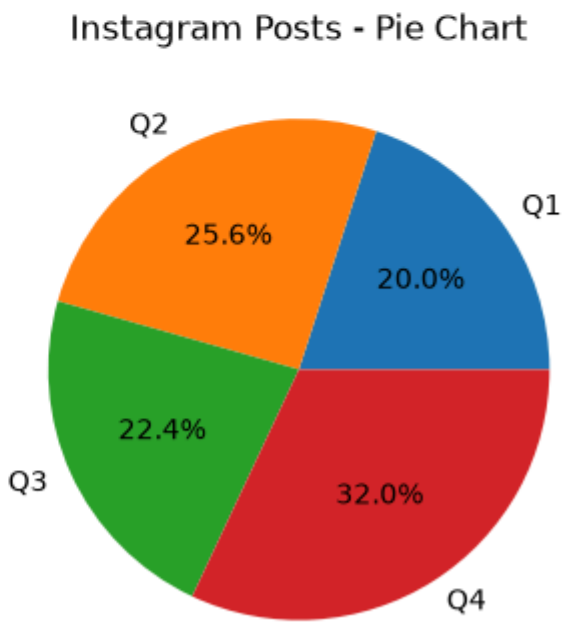
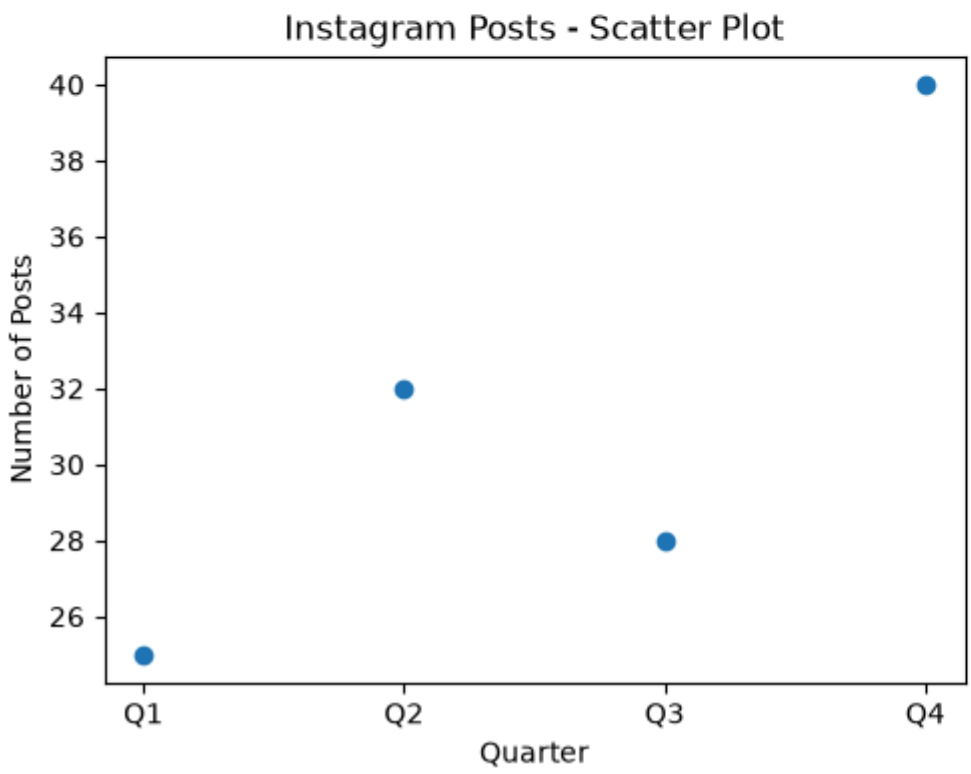
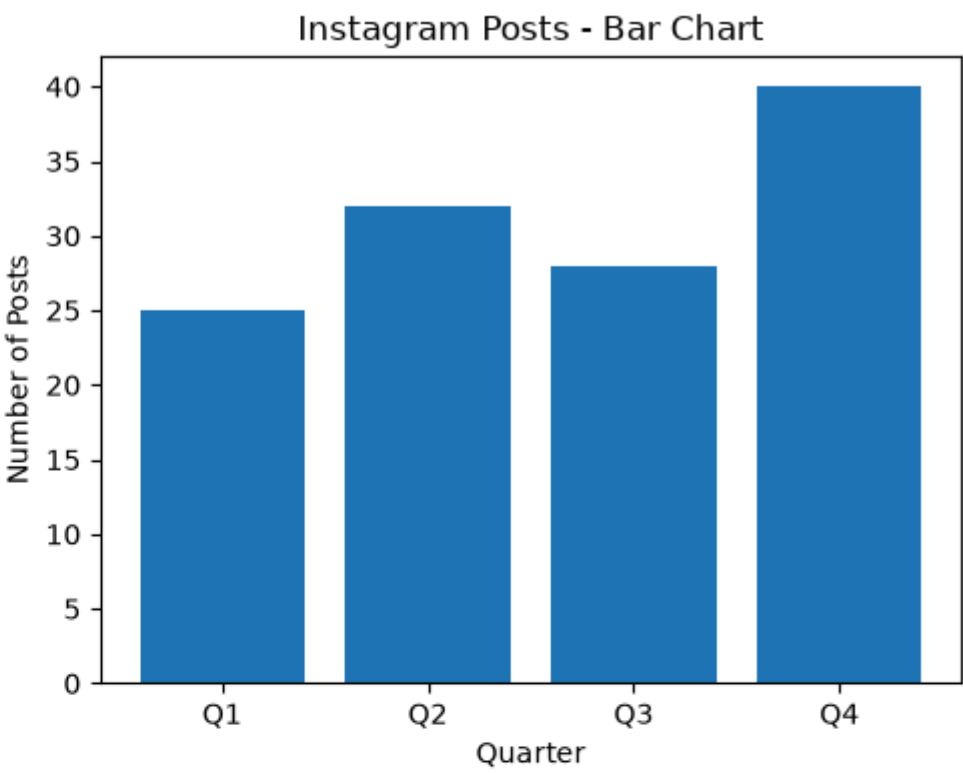
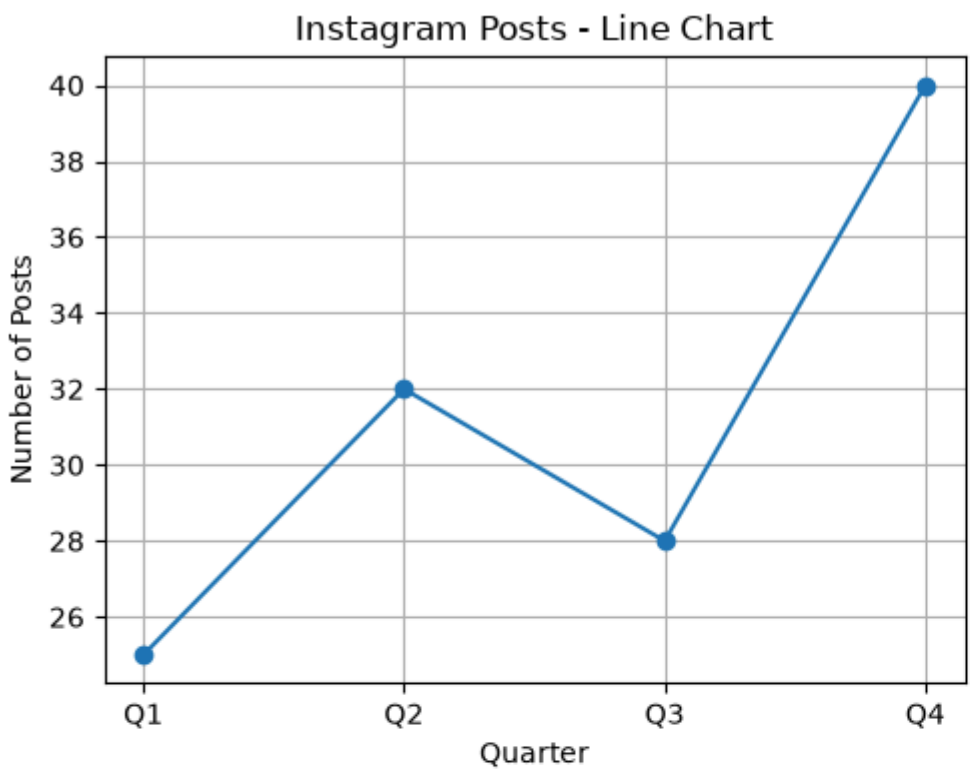
```
ax[0, 1].bar(quarters, posts)
ax[0, 1].set_title("Instagram Posts - Bar Chart")
ax[0, 1].set_xlabel("Quarter")
ax[0, 1].set_ylabel("Number of Posts")
```

```
# Q3 - Scatter plot
ax[1, 0].scatter(quarters, posts)
ax[1, 0].set_title("Instagram Posts - Scatter Plot")
ax[1, 0].set_xlabel("Quarter")
ax[1, 0].set_ylabel("Number of Posts")
```

```
# Q4 - Pie chart
ax[1, 1].pie(posts, labels=quarters, autopct="%1.1f%%")
ax[1, 1].set_title("Instagram Posts - Pie Chart")
```

```
plt.tight_layout()
```

```
plt.show()
```



```
In [7]: #Task 3 – Zomato, Swiggy & Domino's

months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]

zomato_orders = [8, 10, 12, 9, 14, 16]
swiggy_orders = [6, 9, 11, 13, 15, 17]
dominos_orders = [4, 6, 5, 8, 9, 10]

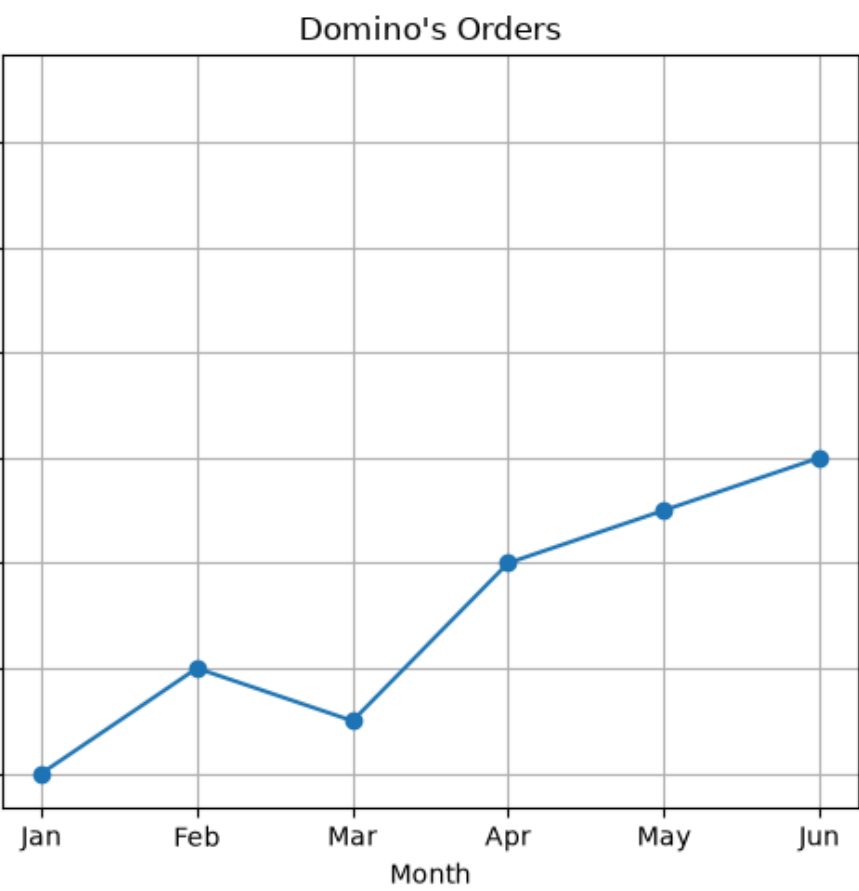
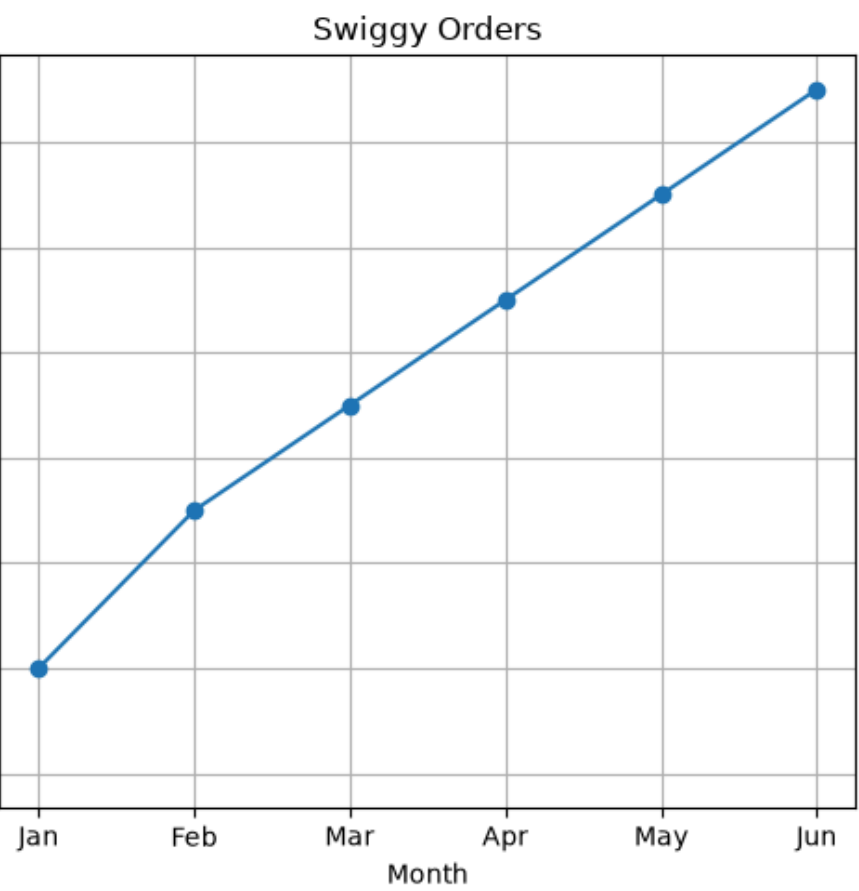
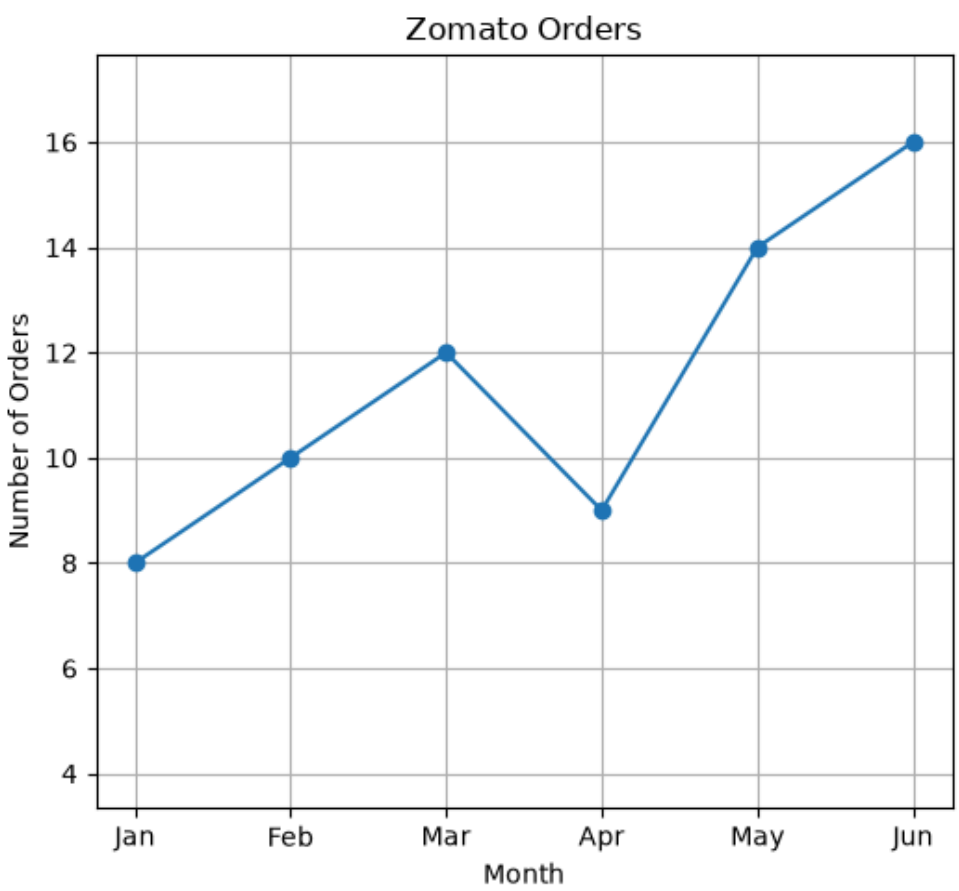
fig, ax = plt.subplots(1, 3, figsize=(15, 5), sharey=True)

ax[0].plot(months, zomato_orders, marker="o")
ax[0].set_title("Zomato Orders")
ax[0].set_xlabel("Month")
ax[0].set_ylabel("Number of Orders")
ax[0].grid()

ax[1].plot(months, swiggy_orders, marker="o")
ax[1].set_title("Swiggy Orders")
ax[1].set_xlabel("Month")
ax[1].grid()

ax[2].plot(months, dominos_orders, marker="o")
ax[2].set_title("Domino's Orders")
ax[2].set_xlabel("Month")
ax[2].grid()

plt.tight_layout()
plt.show()
```



```
In [9]: #Task 4 – Mobile Data Usage

months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
youtube = [12, 15, 18, 20, 22, 25]
instagram = [8, 10, 12, 11, 14, 16]
spotify = [5, 6, 7, 8, 9, 10]
whatsapp = [3, 4, 5, 4, 6, 7]

fig, ax = plt.subplots(2, 2, figsize=(10, 8), sharex=True)

ax[0, 0].plot(months, youtube, marker="o")
ax[0, 0].set_title("YouTube Data Usage")
ax[0, 0].set_ylabel("Data Used (GB)")
ax[0, 0].grid()

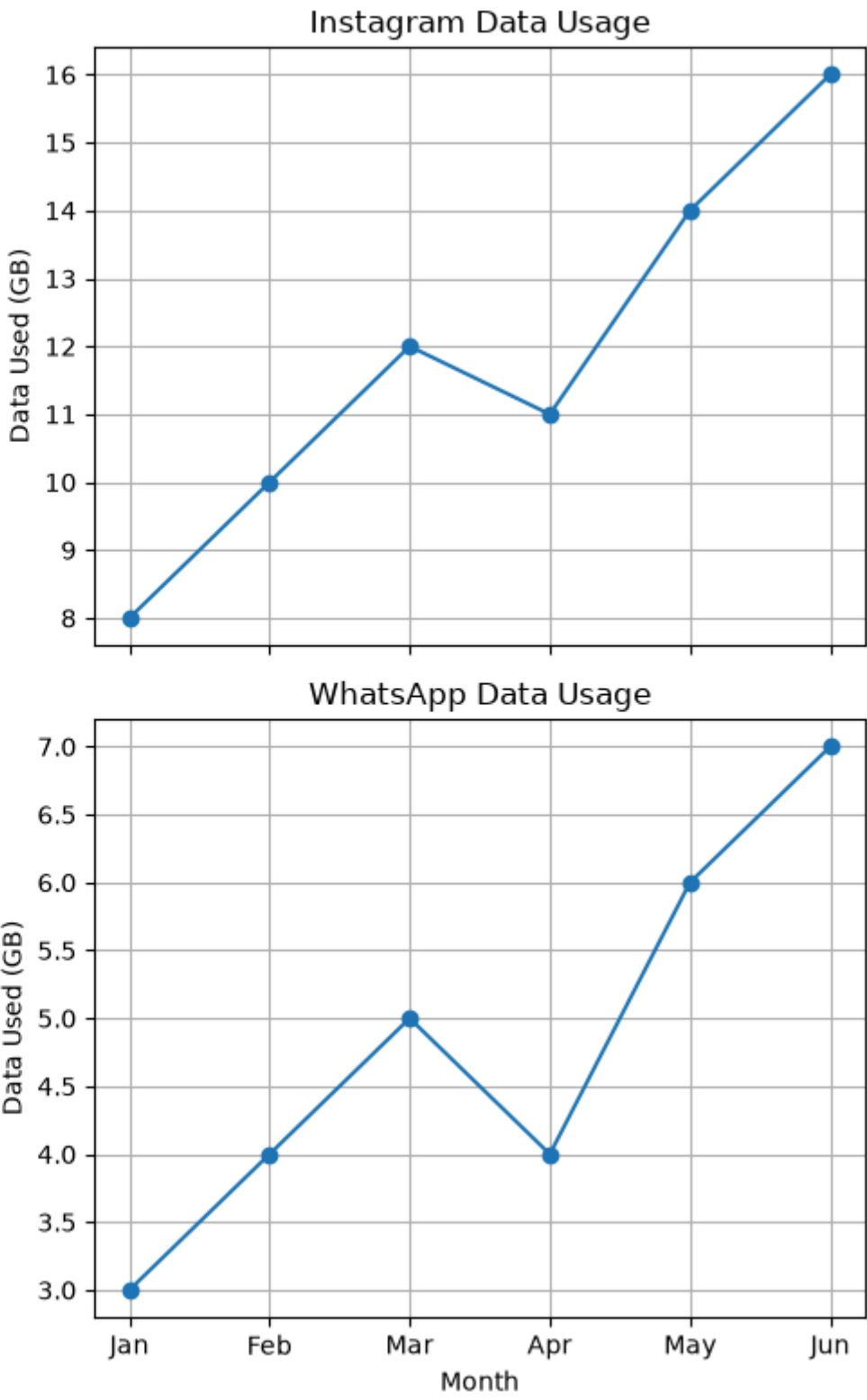
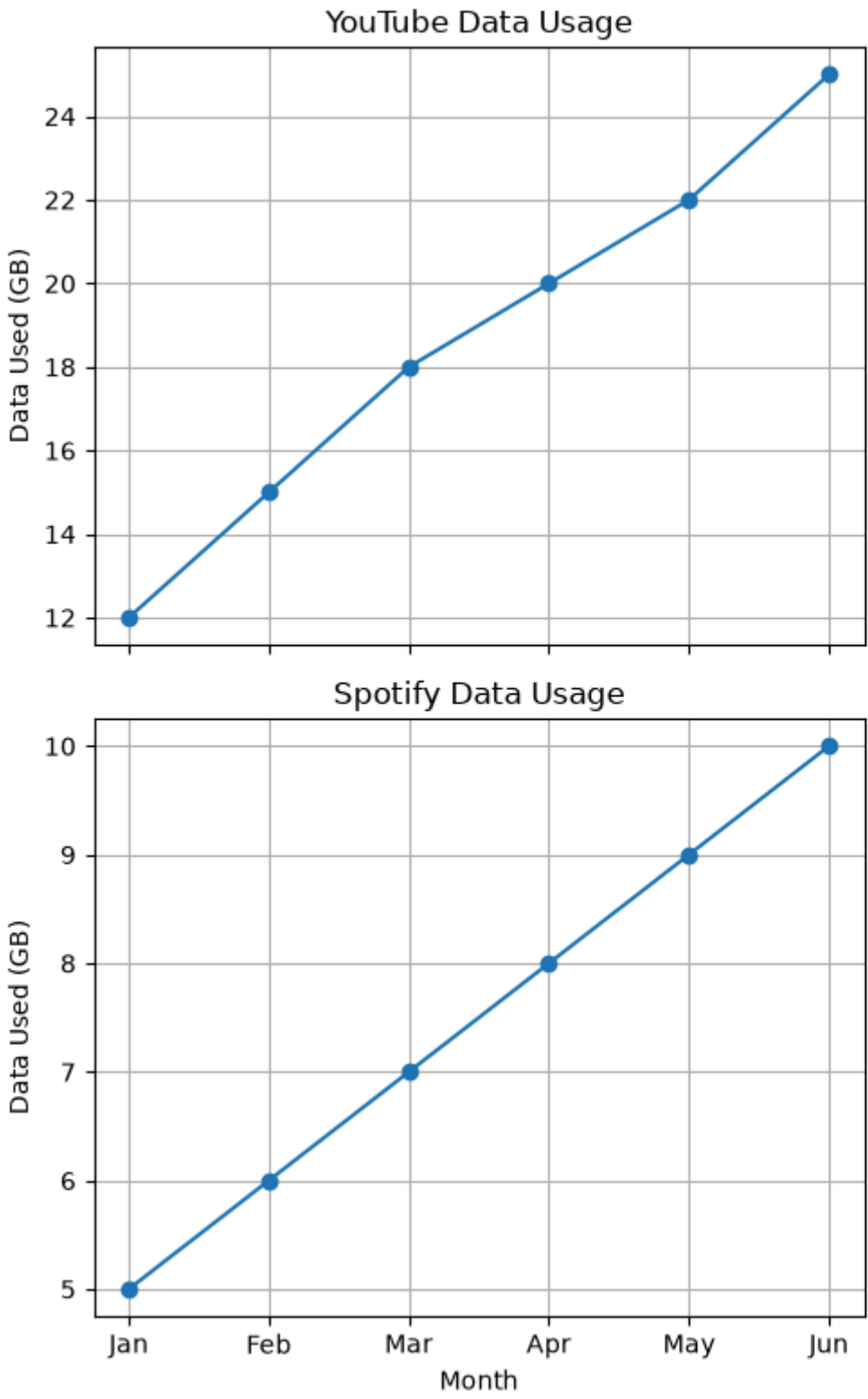
ax[0, 1].plot(months, instagram, marker="o")
ax[0, 1].set_title("Instagram Data Usage")
ax[0, 1].set_ylabel("Data Used (GB)")
ax[0, 1].grid()

ax[1, 0].plot(months, spotify, marker="o")
ax[1, 0].set_title("Spotify Data Usage")
ax[1, 0].set_xlabel("Month")
```

```
ax[1, 0].set_ylabel("Data Used (GB)")
ax[1, 0].grid()

ax[1, 1].plot(months, whatsapp, marker="o")
ax[1, 1].set_title("WhatsApp Data Usage")
ax[1, 1].set_xlabel("Month")
ax[1, 1].set_ylabel("Data Used (GB)")
ax[1, 1].grid()

plt.tight_layout()
plt.show()
```



In []: